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Guidelines for assessing postnatal problems: introducing evidence-based guidelines in Australian general practice

Jane Gunn, Donna Southern, Patty Chondros, Philippa Thomson and Kathryn Robertson

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Background. Postnatal morbidity is high, and many GPs lack the confidence and knowledge to deal with common postnatal problems. There is a high consultation rate, but few women disclose common health problems.

Objective. The aim of the present study was to increase the knowledge and skills of GPs to enable them to identify and manage common health problems experienced by women in the year following childbirth.

Methods. An educational programme [Guidelines for Assessing Postnatal Problems (GAPP)] embedded within a large randomized community intervention trial [Program of Resources, Information and Support for Mothers (PRISM)] with a before/after evaluation was undergone by Australian GPs working in four metropolitan and four rural communities. The programme comprised audit, interactive workshops, role-play and evidence-based guidelines, and was evaluated at baseline and 6 months through written questionnaires and a surgery consultation with a trained simulated patient evaluator.

Results. A total of 68 (86%) GPs took part in the full GAPP programme. The odds of a GP improving on the knowledge items ranged from 1.0 to 16, with the greatest change occurring in knowledge about the effectiveness of cognitive behavioural therapy for maternal depression. Of the GPs with an incorrect response at baseline, the percentage demonstrating improved knowledge at follow-up ranged from 22 to 100%. Around half of the GPs demonstrated excellent communication skills at baseline. Of the remaining GPs, more than half demonstrated greatly improved skills to detect common postnatal problems at follow-up. At baseline simulated patient visit, 70% of GPs inquired about sexual problems yet none inquired about the possibility of abuse, whereas at follow-up 94% inquired about sexual problems and 51.5% facilitated the disclosure of physical and emotional abuse. Anonymous feedback on the programme by participating GPs showed that 89% believed the programme positively influenced their actual practice. Interestingly, GPs demonstrated greater knowledge and skills in the simulated setting than on the written questionnaire.

Conclusions. This relatively brief multifaceted educational programme assisted many participants in improving their knowledge and the skills required to improve both physical and emotional health after birth. Despite being experienced clinicians and participating actively in a programme on interviewing skills, half of the GPs did not facilitate disclosure of the underlying sensitive issue (abuse) during the follow-up consultation and could benefit from further in-depth training in effective communication skills.

Keywords. Clinical competence, family practice, maternal depression, patient simulation, postnatal care.

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Department of General Practice, University of Melbourne, 200 Berkeley Street, Carlton, Victoria 3053, Australia. Correspondence to Associate Professor Jane Gunn; E-mail: j.gunn@unimelb.edu.au

Introduction

Pregnancy and birth are major events in women's lives. The last 20 years have seen the systematic documentation of concerning levels of postnatal morbidity in the UK,[1–3] the USA[4] and Australia.[5] There has been little

reduction in the prevalence of common physical problems such as exhaustion, backache, urinary incontinence and perineal pain since the early 1900s,[6] while depression after birth affects between 10 and 20% of women giving birth.[5,7–9] Reducing the severity and duration of depression after birth is a National Health goal in both Australia[10] and the UK.[11] The past decade has seen a resurgence of interest in researching postnatal care and identified that many GPs lack the knowledge and confidence to deal with common postnatal problems[13] and that the traditional 6-week postnatal check-up is not meeting mothers' needs.[12–14] Recent mothers have many contacts with primary health care providers such as GPs[15] and maternal and child health nurses,[16] yet many problems are not disclosed; only one in three women experiencing depression will seek their help,[17] and physical problems often go unreported.[5,18] Facilitating disclosure of depression in primary care is challenging, as practitioners often fail to detect the problem,[19] yet the routine administration of screening questionnaires does not lead to improvements in patient outcomes.[20] A call has been made for care to be based on identifying individual women's needs rather than on a rigid and standardized approach to care.[6]

This paper reports the evaluation of a carefully planned educational intervention aimed at introducing evidence-based guidelines for postnatal care in Australian general practice: Guidelines for Assessing Postnatal Problems (GAPP). GAPP was the separately funded GP educational component of a large 5-year (1997–2002) randomized community intervention trial: Program of Resources, Information and Support for Mothers (PRISM). PRISM aims to improve the physical and emotional health of women following childbirth via an integrated programme of primary care and community-based strategies. Key elements of PRISM include a community development officer, local steering committees, mothers' information kit and vouchers to encourage 'time out' from mothering, befriending opportunities, changes to make local environments more mother (and baby) friendly and educational programmes for primary health care professionals.

Specific aims of GAPP were: to increase GPs' active listening skills as measured by an announced simulated patient evaluator (SPE); to increase the ability of GPs to pick up on cues during a consultation as measured by the SPE; and to increase the knowledge of GPs about the prevalence and management of common postnatal problems as measured by a written questionnaire.

Methods

A multidisciplinary team designed the programme over 6 months to meet the needs of GPs, as determined by statewide surveys of GPs[21] and recent mothers,[5] using current evidence about effective ways to improve clinicians' behaviour.[22–25] A total of 527 GPs were practising in the eight local government areas, yet available funding allowed for a maximum of 100 participants. Limited funding and the evidence that participation rates in individual education programmes are low, typically ~15% of the eligible group,[26] prompted us to include a variety of strategies to increase the reach of the programme. We drew on adult social learning theories[25] in the delivery of the programme and the health programme planning model[27] to implement and sustain the programme. We recruited a local GP advisor in each area to act as 'local GAPP champion', to target GPs caring for large numbers of recent mothers and to maximize recruitment from a wide variety of practices. The GP advisors were equipped with a resource kit and speaker's kit to enable them to promote GAPP messages opportunistically and to visit practices not able to attend the full programme. Regular teleconferences were held with the researchers and the GP advisors.

The full GAPP programme was available to GPs at geographically convenient locations, and included the use of multiple strategies (interactive workshops, role-play, simulated patient clinic visits with confidential feedback, peer discussion groups, evidence-based guidelines, interviewing prompts, clinical audit and programme newsletters). GAPP was promoted as a simple evidence-based approach to maternal health, providing updates on common postnatal physical problems and postnatal depression, as well as focusing on interviewing skills. GPs received a generous allocation of 78 CME (continuing medical education accreditation) points for completing the entire programme. Table 1 shows an outline of the GAPP programme.

An interviewing prompt was designed as a diagnostic aid using the acronym ESP, reminding GPs to ask about emotional, social and physical issues during postnatal consultations. Key question prompts were included under the ESP headings.

Evidence-based guidelines were developed by a multidisciplinary team for common postnatal problems (exhaustion, backache, sexual problems, perineal pain, urinary incontinence, faecal incontinence and depression).[28] The guidelines, attractively presented and laminated, provided a brief update about the condition and listed management strategies with supporting levels of evidence clearly stated.[29] The active listening model used to help GPs diagnose depression was an interactive, engaging process of verbal and non-verbal techniques. GPs were encouraged to use silence, minimal encouragers and to give undivided attention to facilitate picking up on cues. The guidelines included simple 'GAPP strategies' presented by the acronym LAST (listen, acknowledge, suggest time out and tackle issues one by one), based on available evidence about interventions likely to be helpful, such as finding someone to talk to who listens and getting some time out from constant baby care.[30]

GAPP newsletters were attractively presented and designed to reinforce the guidelines and provide useful information.

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TABLE 1 *Outline of the GAPP programme*

Step	Month	Description
Baseline evaluation		
1	1	Workshop 1 (2.5 h)—outline of programme, informed consent to SPE visits, define learning objectives, surveys completed
2	1	Clinical Audit 1—collect data (reason for encounter, problems identified, management initiated) on women seen less than 1 year postpartum, or mothers who bring babies less than 1 year of age
3	1	Simulated Patient Visit 1—consult with 'Chris O'Malley' using a standard consultation of 10–15 min followed by 15 min feedback from SPE, complete brief consultation reflection sheet and refine learning objectives
Introduction of evidence-based guidelines		
4	1	Workshop 2 (4 h)—dissemination and discussion of evidence-based guidelines for postnatal care (including discussion of 'readiness to change behaviour') and GAPP resource folder, small group work on listening skills using role-play and simulated patients, discussion of audit findings, simulated patient visit and revision of learning objectives
5	2–4	Practice-based learning—self-directed using GAPP resource folder, laminated GAPP evidence-based guidelines, GAPP newsletters (sent monthly)
Outcome evaluation		
6	7	Clinical Audit 2—collect data (reason for encounter, problems identified, management initiated) on women seen less than 1 year postpartum, or mothers who bring babies less than 1 year of age
7	7	Simulated Patient Visit 2—consult with 'Chris O'Malley' using a standard consultation of 10–15 min followed by 15 min feedback from SPE, complete brief consultation reflection sheet
8	7	Workshop 3 (2.5 h)—final evaluation session, evidence update

Outcome evaluation

Outcome evaluation was conducted 6 months following the intervention (see Table 1). Ethics approval was obtained from the University of Melbourne's human research ethics committee.

GAPP used a pre-test–post-test design. The main outcomes were pre-specified. Data were obtained from a number of perspectives using written self-report as well as the ratings of SPEs. Self-report was used in three ways: asking GPs about their knowledge, skills and attitudes and to apply this knowledge in the pre-/post-GAPP surveys completed following the workshops, and asking them to list what they had derived from the programme on feedback evaluation forms. GPs' competence to use their knowledge about postnatal problems was assessed by trained SPEs seen by the GP during their usual clinic setting.

GAPP self-report questionnaire

The survey, based on a previous Australian survey of GPs conducted by JG,[21] contained nine sections which covered common physical problems, postnatal depression, general postnatal role, interaction with maternal and child health nurses, interaction with other health professionals about maternal health, attitudes towards postnatal care, reflection about own practice, communication skills and demographic characteristics. Survey items included 5-point Likert scales, specified categories and open-ended items. The survey was piloted using a convenience sample of six GPs (male, female, rural, urban, high and low levels of interest in postnatal care) and the wider PRISM research team.

A coding system for open-ended questions was developed (JG and DS) drawing on the evidence-based information about maternal health problems and treatment strategies included in the GAPP guidelines. Each response was read and discussed by JG and DS and coded according to whether the GP had incorporated the key components in their response. Data were entered onto an Excel database and analysed using STATA 6.0.[31]

Simulated patient evaluators (SPEs)

Consenting GPs were booked to conduct a standard consultation (usually 15 min) with 'Chris O'Malley' (trained SPE), who presented at baseline as 3 months postpartum with recurrent upper respiratory tract infections, tiredness and exhaustion with the underlying issue of sexual problems. GPs were aware, at the time of the consultation, that 'Chris O'Malley' was an SPE from the GAPP project. The same SPE returned at follow-up as 'Chris O'Malley' 9 months postpartum with backache and the underlying issue of abuse. The SPEs participated in the consultation, then left the room for 5 min to complete a rating scale (communication and case-specific items) and then returned to give feedback to the GP for ~15 min.

The four SPEs had been trained during an intensive 30 h training programme.[32] Intra-rater reliability was

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high (median kappa score range 0.8–0.87); inter-rater reliability was moderate to almost perfect (percentage concordance range 62–68; kappa summary statistic 0.46–0.95).

Sample size calculations were based on McNemar's chi-squared test for paired proportions to test differences pre- and post-intervention. We considered an effect size of 15–20% (power = 80%, alpha < 0.05) of GPs changing their response for selected items in the direction anticipated after the educational intervention to be the minimum required to justify the expense and effort of conducting the programme.

Demographic and GP feedback data were summarized with frequencies and percentages. Most ordinal responses were collapsed to a binary response. The analysis takes into account the matching of pre- and post-responses of GPs. Differences in the paired proportions were tested using McNemar's chi-squared statistic. Odds ratios (ORs) were calculated using the ratio of the discordant pairs and reported with the respective 95% confidence intervals (CIs). GP knowledge was assessed using ORs of improvement, i.e. the odds of a GP changing from an incorrect response at baseline to a correct response at 6-month follow-up compared with the odds of changing from a correct response at baseline to an incorrect response. It should be noted that GPs who gave the same response at baseline and follow-up did not contribute any information to the hypothesis testing. Paired ordinal variables were summarized using frequencies, and differences were tested using Wilcoxon matched pairs signed-ranks test.

Results

Eight GP advisors were recruited and remained active throughout the programme. All local GPs (527) received the laminated GAPP guidelines and three GAPP newsletters. Thirteen practices (32 GPs) received a tailored visit from the GP advisor. GAPP advisors hosted three separate seminars for local GPs unable to attend the full GAPP programme. Eighty-one GPs registered for the full GAPP programme and 79 attended the first workshop. Eleven withdrew from the programme before the outcome evaluation at 6 months. Reasons for withdrawal were lack of time (seven), moved away (two), maternity leave (one) and reached required quota for CME (one). Sixty-eight participants took part in both simulated patient visits, and 61 (90%) provided complete data for pre- and post-written surveys. This paper reports data from the 68 participants who completed the full GAPP programme and both SPE visits.

Thirty-eight (56%) GPs were practising in a Metropolitan area and 30 (44%) in a rural area. Thirty-nine GPs (57%) were female and 29 (43%) were male. Most GPs (87%) were aged 31–50 years and 85% reported having their own children. More than half of the GPs (54%) held a Diploma of Obstetrics, 40% held a FRACGP and 25% held no qualification other than MBBS.

Detailed information was available from responses to an open-ended question about common physical health problems. The three most common physical problems in the first year after giving birth are exhaustion, backache and sexual problems.[5] GPs nominating these rose from 61% (37/61) to 71% (43/61) for exhaustion, from 25% (41/61) to 71% (43/61) for backache and from 33% (20/61) to 49% (30/61) for sexual problems.

Table 2 presents the paired data analysis for the GPs' self-reported knowledge. Of those with an incorrect response at baseline, the percentage demonstrating improved knowledge at follow-up ranged from 22 to 100%. By adding the first (correct–correct) and third column (correct–incorrect), the proportion of GPs with a correct response at baseline is obtained, and by adding the first two columns (correct–correct and incorrect–correct), the proportion of GPs with a correct response after the intervention is obtained.

Responses to an open-ended question about usual treatment for maternal depression showed that GPs reporting the key GAPP messages as management strategies rose from 16% (10/61) to 49% (30/61) (OR = 6.0; 95% CI 2.0–24) for active listening and from 13% (8/61) to 38% (23/61) (OR = 6.0; 95% CI 1.8–32) for time out.

Table 3 provides details of the health problems mentioned and discussed during the SPE consultations. GPs were more likely to cover the sensitive issues of sex, incontinence and bowel problems at the second SPE visit. The proportion of GPs mentioning and discussing these issues at baseline can be calculated by adding the yes–yes column and the yes–no column in Table 3. Similarly, the proportion of GPs mentioning and discussing these issues after the intervention can be calculated by adding the yes–yes column and no–yes column.

Due to the focus in GAPP on depression, participants were likely to have heightened awareness of potential maternal depression so, instead of incorporating maternal depression into the SPE case, we decided to measure the GPs' ability to facilitate disclosure of sensitive issues using sexual problems (at baseline) and partner abuse (at follow-up). At baseline, 70% of GPs inquired about sexual problems yet none inquired about the possibility of abuse. At follow-up, 94% inquired about sexual problems and 51.5% facilitated the disclosure of physical and emotional abuse (see Table 3).

Sixty-eight GPs participated in a consultation at their usual practice, during normal clinic time with a trained SPE at both baseline and 6-month follow-up. Prior to data analyses, five out of the 15 items on the communication skills rating scale were identified as related to 'active listening'. Table 4 shows these items and presents the paired data analysis. On completion of follow-up, no doctor seemed rushed, hence it was not possible to calculate an OR for this item. Of the remaining four items, ORs were obtained for improved skills, with twice as many GPs likely to improve their skills rather than

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TABLE 2 *Self-reported knowledge about postnatal health problems pre- and post- the GAPP intervention (presented as paired data)*

Item	True/ False	Pre–post response (n =60)[b]				OR	95% CI
		Correct–correct n (%)	Incorrect–correct n (%)	Correct–incorrect n (%)	Incorrect–incorrect n (%)		
Women who have an operative delivery are more likely to experience problems with fatigue/tiredness in the year after birth than women who have spontaneous vaginal birth	T	19 (31.7)	12 (20)	6 (10)	23 (38.3)	2.0	(0.7–6.5)
Backache is one of the commonest physical health problems a recent mother may experience	T	49 (81.7)	10 (16.7)	0	1 (1.7)	–	
Women who experience a problem with perineal pain are likely to raise this issue with their GP[a]	F	29 (49.2)	13 (22)	5 (8.5)	12 (20.3)	2.6	(0.9–9.3)
Therapeutic ultrasound is an effective treatment for perineal pain	F	8 (13.3)	11 (18.3)	2 (3.3)	39 (65)	5.5	(1.2–51)
Women with stress incontinence are likely to benefit from pelvic floor exercises	T	59 (98.3)	1 (1.7)	0	0	–	
Routine vaginal examination at the 6 week check/routine postnatal review will seldom reveal a problem that needs to be addressed	T	29 (48.3)	6 (10)	6 (10)	19 (31.7)	1.0	(0.3–3.7)
Having a supportive partner makes a difference to the likelihood of experiencing maternal depression after birth	T	52 (86.7)	5 (8.3)	3 (5)	0	1.7	(0.32–11)
Maternal depression in the year after birth should usually by managed by a psychiatrist expert in the area[a]	F	51 (86.4)	5 (8.5)	0	3 (5.1)	–	
Simple techniques such as active listening and non-directive counselling are relatively ineffective strategies in reducing maternal depression	F	51 (85)	8 (13.3)	1 (1.7)	0	8	(1.1–355)
Women who breastfeed are more likely to experience maternal depression in the year after birth[a]	F	31 (52.5)	13 (22)	10 (16.9)	5 (8.5)	1.3	(0.53–3.3)
Many cases of maternal depression in the postnatal care period are caused by hormonal changes after the birth	F	29 (48.3)	11 (18.3)	4 (6.7)	16 (26.7)	2.8	(0.81–12)
Cognitive behavioural therapy is as effective as fluoxetine in reducing maternal depression[a]	T	23 (39)	16 (27.1)	1 (1.7)	19 (32.2)	16	(2.5–671)
Three-quarters of recent mothers who report exhaustion will also be depressed[a]	F	11 (18.6)	10 (16.9)	5 (8.5)	33 (55.9)	2.0	(0.62–7.5)

[a] n = 59.
[b] Missing data for one GP, thus sample size = 60.

demonstrate poorer communication skills. However, due to the relatively small numbers of discordant pairs, statistical significance was obtained for only one item ("Did you get a chance to say what was really on your mind?").

Of the 54 (79%) GPs completing the anonymous evaluation on completion of the programme, 25 (46%) volunteered the simulated patient visit to their clinic as the most useful element of the programme and 17 (32%) rated the visit as the most challenging aspect of the programme. Role-play and listening skills were mentioned by 11 GPs (20%). Eighty-nine percent reported that the programme had positively influenced their practice, while two (3.7%) were unsure and four (7.4%) reported no change. The most commonly reported outcomes of participation were: increased listening 14 (25.9%), asking more questions 10 (18.5%) and increased awareness of unspoken problems nine (16.7%).

Discussion

GAPP focused on common postnatal problems to provide GPs with an opportunity to gain the knowledge and skills necessary to provide evidence-based postnatal care. Combining physical and mental health issues integrated with a holistic patient-centred interviewing approach into the one programme required a balance between the amount of material presented and the way GPs work in the clinical setting. The outcome was an emphasis on key programme messages rather than an in-depth focus on the specific aspects of management of each condition, which instead was available in the laminated guidelines.

Participants came from a variety of metropolitan and rural settings, and female practitioners were over-represented, which is in line with the finding that female GPs are more likely to provide postnatal care.[21] GPs

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TABLE 3 *Health issues mentioned and discussed by the GP during baseline and follow-up consultations with 'Chris O'Malley', as rated by the simulated patient evaluator*

Case-specific schedule	Pre–post (n = 68)				OR	95% CI
	Yes–yes n (%)	No–yes n (%)	Yes–no n (%)	No–no n (%)		
Feeling tired and exhausted[a]	65 (95.6)	0	3 (4.4)	0	–	(0–2.4)
Coughs, colds and other minor illnesses[a]	90 (13.2)	58	1 (85.3)	– (1.5)	(0–0.07)	
Your sleep	59 (86.8)	0	9 (13.2)	0	–	(0–0.51)
Backache[b]	19 (27.9)	48 (70.6)	0	1 (1.5)		
Sex	46 (67.6)	18 (26.5)	2 (2.9)	2 (2.9)	9.0	(2.2–80)
Haemorrhoids	0	3 (4.4)	13 (19.1)	52 (76.5)	0.23	(0.04–0.84)
Perineal pain	2 (2.9)	6 (8.8)	7 (10.3)	53 (77.9)	0.86	(0.24–3.0)
Breastfeeding	66 (97.1)	1 (1.5)	1 (1.5)	0	1.0	(0.01–78)
Bowel problems	6 (8.8)	18 (26.5)	6 (8.8)	38 (55.9)	3.0	(1.14–9.2)
Incontinence	8 (11.8)	25 (36.8)	7 (10.3)	28 (41.2)	3.6	(1.5–9.8)
Not coping	62 (91.2)	1 (1.5)	4 (5.9)	1 (1.5)	0.25	(0.01–2.5)
Sex	46 (67.6)	18 (26.5)	2 (2.9)	2 (2.9)	9.0	(2.2–80)
Sexual, physical or emotional abuse	0	35 (51.5)	0 (48.5)	33		

[a] Feeling tired and exhausted, and coughs, colds and other minor illnesses were the presenting complaints at baseline.
[b] Backache was the presenting complaint at follow-up.

TABLE 4 *Simulated patient evaluation of GP active listening skills*

Communication skills schedule	Pre–post (n = 68) (%)				OR	95% CI	P^a
	Yes–yes	No–yes	Yes–no	No–no			
Did the doctor make you feel that you could talk about any type of problem?	27 (39.7)	18 (26.5)	8 (11.8)	15 (22)	2.3	(0.93–6.0)	0.08
Did you get a chance to say what was really on your mind?	25 (36.8)	19 (27.9)	7 (10.3)	17 (25)	2.7	(1.1–7.6)	0.03
Did the doctor give you undivided attention?	47 (69.1)	12 (17.6)	5 (7.3)	4 (5.9)	2.4	(0.8–8.7)	0.14
Did the doctor explore the cues you dropped, with you?	18 (26.5)	20 (29.4)	9 (13.2)	21 (30.9)	2.2	(0.97–5.5)	0.06
Did this doctor seem rushed?	0	0 (2.9)	2 (97)	66	–	(0–5.3)	0.50

[a] McNemar's test.
OR = matched odds ratio.

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participating in GAPP were self-selecting, and the data presented here represent the knowledge and skills of an interested group, yet still one that could benefit from improved knowledge and skills in postnatal care.

Whilst substantial evidence exists to support multifaceted educational programmes, there is a lack of evidence on the most cost-effective implementation strategies. The conduct of GAPP within a larger community intervention trial is likely to have raised GP awareness about the programme and to have enhanced recruitment. Despite this, the multifaceted nature of the programme and incorporation of local GP advisors into the study team, participation rates remained similar to those reported in other published studies.[26] It is likely that more GPs would have taken part in the full GAPP programme if we had been able to offer them a choice of workshop dates; however, budget constraints prevented this. Research is required to find cost-effective ways to maximize participation rates in educational programmes.

GAPP participants were encouraged to use active listening and a simple interview prompt (ESP) to pick up cues in order to facilitate disclosure of depression and other sensitive issues, followed by use of effective communication skills and evidence-based guidelines to discuss a management plan with the woman. The pre–post design, use of announced SPEs and different scenarios at baseline and follow-up limit the extent to which conclusions about causation of any identified changes can be made. However, many GPs demonstrated improved knowledge and clinical competence to detect and manage sensitive issues. A challenging aspect of GAPP was the introduction of evidence that questioned some GPs' firmly held beliefs. Around half the GPs expressed surprise that evidence was lacking for a hormonal basis of maternal depression and that psychological and drug therapies were equally effective management strategies. Between a third and half of these GPs were willing to accept the evidence and change their minds. We were unable to find published comparative data about uptake of evidence that challenges firmly held beliefs.

At baseline, the range of physical problems listed by GPs was broad and did not reflect the actual prevalence of postnatal problems. By follow-up, the problems nominated reflected their actual prevalence in the community. Significant improvements were made in knowledge about the effectiveness of non-drug treatments for depression. One-third of the GPs remained convinced that hormonal changes caused many cases of maternal depression, despite the programme discussions and guidelines indicating a lack of evidence to support this belief.

Despite being experienced clinicians and participating actively in a programme on interviewing skills, half of the GPs did not facilitate disclosure of the underlying sensitive issue (abuse) during the follow-up consultation. Many GPs at the final programme workshop commented that they were amazed they had 'missed it' (abuse) and were intrigued as to how their colleagues managed to facilitate disclosure of this issue. To explore our findings further, we are following-up a subgroup of the GPs who took part in GAPP and collecting information about disclosure of sensitive issues from 50 consecutive female patients.

Particularly in the area of physical health, GPs were more likely to demonstrate better knowledge in the active simulated situation than in the written questionnaire. For example, exhaustion (the most common physical postnatal problem) was discussed in 95% of simulated patient encounters, but only 61% included this as a common problem on the written questionnaire. General practice is very much a consultation-driven, patient-centred discipline, and researchers should consider this when evaluating educational programmes, as written instruments alone may fail to detect important changes.

GAPP is unique in that it was conducted as a GP educational programme within a large community intervention trial (PRISM) designed to improve maternal health. During 2003, health outcome data for women giving birth in PRISM intervention and comparison communities and information about their experience of GP services will be available.

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References

[1] MacArthur C, Lewis M, Knox EG. Health after childbirth. *Br J Obstet Gynaecol* 1991; **98:** 1193–1204.

[2] MacArthur C, Lewis M, Knox E. *Health after Childbirth*. London: HMSO; 1991.

[3] Bick DE, MacArthur C. The extent, severity and effect of health problems after childbirth. *Br J Midwifery* 1995; **3:** 27–31.

[4] Glazener C, Abdalla M, Stroud P, Naji S, Templeton A, Russell I. Postnatal maternal morbidity: extent, causes, prevention and treatment. *Br J Obstet Gynaecol* 1995; **102:** 282–287.

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[5] Brown S, Lumley J. Maternal health after childbirth: results of an Australian population based survey. *Br J Obstet Gynaecol* 1998; **105:** 156–161.

[6] MacArthur C. What does postnatal care do for women's health? *Lancet* 1999; **353:** 343–344.

[7] Astbury J, Brown S, Lumley J, Small R. Birth events, birth experiences and social differences in postnatal depression. *Aust J Publ Health* 1994; **18:** 176–184.

[8] Kumar R, Robson KM. A prospective study of emotional disorders in childbearing women. *Br J Psychiatry* 1984; **144:** 35–47.

[9] Brown S, Lumley J. Physical health problems after childbirth and maternal depression at six to seven months postpartum. *Br J Obstet Gynaecol* 2000; **107:** 1194–1201.

[10] Nutbeam D, Wise M, Bauman A, Harris E *et al*. *Goals and Targets for Australia's Health in the Year 2000 and Beyond*. Canberra: Australian Government Printing Service; 1993.

[11] Department of Health. *A National Service Framework for Mental Health*. London: DoH; 1999.

[12] Gunn J. The six week postnatal check up. Should we forget it? *Aust Fam Physician* 1998; **27:** 399–403.

[13] Noble T. The routine six week postnatal vaginal examination. *Br Med J* 1993; **307:** 698.

[14] Sharif K, Clarke P, Whittle M. Routine six-week postnatal vaginal examination—to do or not to do? *J Obstet Gynaecol* 1993; **13:** 251–252.

[15] Gunn J, Lumley J, Young D. Visits to medical practitioners in the first 6 months of life. *Paediatr Child Health* 1996; **32:** 162–166.

[16] Keys-Brown J. Maternal and child health visits in the first two years of life in Prahran, Victoria. Thesis, Master of Public Health. Melbourne: Monash University; 1993.

[17] Small R, Brown S, Lumley J, Astbury J. Missing Voices: what women say and do about depression after childbirth. *J Reprod Infant Psychol* 1994; **12:** 89–103.

[18] Glazener C, MacArthur C, Garcia J. Postnatal care: time for a change. *Contemp Rev Obstet Gynaecol* 1993; **5:** 130–136.

[19] Regier D, Hirshfeld R, Goodwin F, Burke J, Lazar J, Judd L. The NIMH depression awareness, recognition and treatment program: structure, aims and scientific basis. *Am J Psychiatry* 1988; **145:** 1351–1357.

[20] Gilbody SM, House AO, Sheldon TA. Routinely administered questionnaires for depression and anxiety: systematic review. *Br Med J* 2001; **322:** 406–409.

[21] Gunn J, Lumley J, Young D. The role of the general practitioner in postnatal care: a survey from Australian general practice. *Br J Gen Pract* 1998; **48:** 1570–1574.

[22] Davis DA, Thomson MA, Oxman AD, Haynes B. Evidence for the effectiveness of CME. A review of 50 randomized controlled trials. *J Am Med Assoc* 1992; **268:** 1111–1117.

[23] Davis D, Fox R. *The Physician as Learner: Linking Practice to Research*. Chicago: American Medical Association; 1994.

[24] Davis D, Thomson M, Oxman A, Haynes R. Changing physician performance. A systematic review of the effect of continuing medical education strategies. *J Am Med Assoc* 1995; **274:** 700–705.

[25] Moulding NT, Silagy CA, Weller DP. A framework for effective management of change in clinical practice: dissemination and implementation of clinical practice guidelines. *Qual Health Care* 1999; **8:** 177–183.

[26] Hulsman RL, Ros WJG, Winnubst JAM, Bensing JM. Teaching clinically experienced physicians communication skills. A review of evaluation studies. *Med Educ* 1999; **33:** 655–668.

[27] Hawe P, Degeling D, Hall J. *Evaluating Health Promotion*. NSW: MacLennan and Petty Ltd; 1995.

[28] Gunn J, Brown S, Small R, Lumley J. *GAPP Evidence Based Guidelines*. Melbourne: Department of General Practice, University of Melbourne; 1999.

[29] NHMRC. *A Guide to the Development, Implementation and Evaluation of Clinical Practice Guidelines*. Canberra: NHMRC; 1998.

[30] Brown S, Lumley J, Small R, Astbury J. *Missing Voices*. Melbourne: Oxford University Press; 1994.

[31] STATA C. Stata for Windows. In *Stata Release Version 6*. College Station Texas: Stata Press; 1999.

[32] Gunn J, Southern D, Chondros P *et al*. *Guidelines for Assessing Postnatal Problems (GAPP): Final Report*. Carlton, Victoria: Department of General Practice and Public Health, The University of Melbourne; 2000.

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